

GLUECKSBURG-MEIERWIK, Germany

WMO: 100330

Lat: 54.8272N

Lon: 9.5058E

Elev: 33

StdP: 100.93

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-4.9	-9.2	1.7	-5.2	-7.6	2.0	-3.8	19.5	6.9	17.7	6.7	3.7	120	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	26.1	19.3	24.4	18.6	22.8	17.7	20.3	24.6	19.3	23.1	18.3	21.8	4.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.7	22.6	17.8	12.8	21.2	16.9	12.1	20.2	58.3	24.7	54.8	22.9	51.8	21.6	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	13.0	11.3	-8.4	28.8	1.8	1.9	-9.8	30.1	-10.8	31.2	-11.9	32.2	-13.2	33.6		
(5)			-8.9	22.1	1.8	0.8	-10.2	22.7	-11.2	23.2	-12.3	23.7	-13.6	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	1.8	2.1	4.0	7.9	12.0	14.9	17.4	17.5	14.4	10.3	5.9	3.1		
	DBStd	6.42	3.61	3.40	3.09	2.97	2.93	2.67	2.78	2.57	2.33	2.89	3.09	3.64		
	HDD10.0	1122	254	222	186	76	13	0	0	0	1	32	125	213		
	HDD18.3	3343	513	454	444	312	198	108	53	48	119	250	372	471		
	CDD10.0	872	0	0	1	14	74	147	229	232	132	41	3	0		
	CDD18.3	53	0	0	0	0	1	5	24	22	2	0	0	0		
	CDH23.3	264	0	0	0	1	4	31	132	94	3	0	0	0		
	CDH26.7	29	0	0	0	0	0	3	14	12	0	0	0	0		
(14) Wind	WSAvg	5.4	6.2	6.1	5.9	5.0	5.0	5.3	4.9	4.7	5.0	5.3	5.4	5.8		
(15) Precipitation	PrecAvg	805	74	56	47	39	52	67	84	87	77	80	67	75		
	PrecMax	1029	158	128	111	114	91	140	168	215	146	170	160	162		
	PrecMin	494	1	10	6	3	14	0	24	35	11	26	11	29		
	PrecStd	125	38	24	25	23	24	32	40	40	37	27	38	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	10.9	14.2	19.1	23.8	26.2	28.2	28.2	23.4	18.9	14.1	11.8		
		MCWB	9.0	7.7	9.6	12.8	17.1	18.7	20.1	19.8	18.4	15.7	12.6	10.4		
	2%	DB	8.7	9.0	11.2	16.9	21.2	24.0	26.6	25.9	21.3	16.8	12.6	10.1		
		MCWB	7.9	7.5	8.1	11.6	15.6	18.2	19.5	19.6	17.0	14.5	11.4	9.3		
	5%	DB	7.4	7.7	9.7	14.9	19.2	21.8	24.9	24.0	19.8	15.2	11.2	9.0		
		MCWB	6.7	6.7	7.4	10.6	14.0	16.6	18.8	18.6	16.1	13.5	10.0	8.3		
	10%	DB	6.6	6.8	8.2	13.0	17.4	19.9	22.9	22.1	18.2	14.2	10.1	8.0		
		MCWB	5.9	5.8	6.6	9.7	13.1	15.6	17.9	17.8	15.2	12.6	9.1	7.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	8.7	10.3	13.9	18.0	20.3	21.7	21.6	19.0	15.9	13.1	10.7		
		MCDB	9.7	9.7	12.6	18.0	22.5	24.9	26.3	25.6	22.1	17.9	14.1	11.5		
	2%	WB	7.8	7.6	8.8	12.3	16.2	18.7	20.5	20.2	17.6	14.9	11.6	9.4		
		MCDB	8.3	8.4	10.6	15.8	19.9	22.8	25.1	24.1	20.4	16.3	12.5	9.9		
	5%	WB	6.7	6.6	7.7	11.1	14.9	17.3	19.4	19.2	16.6	13.8	10.3	8.2		
		MCDB	7.2	7.4	9.0	14.1	18.3	20.6	23.4	22.8	18.9	15.1	11.1	8.7		
	10%	WB	5.8	5.7	6.8	9.9	13.6	16.1	18.3	18.3	15.7	12.9	9.2	7.2		
		MCDB	6.4	6.5	8.0	12.5	16.7	19.2	22.0	21.4	17.7	14.2	9.9	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	4.0	5.3	7.2	7.7	7.3	7.4	7.2	6.1	5.0	4.0	3.6		
		MCDBR	4.0	4.7	7.8	10.2	11.0	10.7	10.6	10.0	8.2	5.8	4.3	3.9		
	5% WB	MCWBR	3.8	3.9	5.2	6.0	6.3	6.0	5.2	5.1	4.8	4.3	3.8	3.9		
		MCWBR	3.9	4.3	6.7	9.1	9.8	9.4	9.5	8.8	7.0	5.4	4.2	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.322	0.341	0.368	0.400	0.393	0.389	0.406	0.405	0.382	0.359	0.337	0.310		
		taud	2.270	2.297	2.262	2.223	2.307	2.342	2.313	2.328	2.374	2.389	2.329	2.265		
	Ebn at Noon		600	731	796	820	851	857	833	809	777	704	580	526		
		Edh at Noon	59	80	103	123	119	117	118	110	92	73	56	48		
(44) All-Sky Solar Radiation	RadAvg		0.51	1.14	2.44	4.07	5.21	5.44	5.10	4.12	2.91	1.54	0.63	0.37		
	RadStd		0.07	0.16	0.22	0.46	0.51	0.44	0.58	0.31	0.31	0.18	0.08	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	-129	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-148	N/A	N/A

Nomenclature: See separate page